

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution Affiliated to Anna University)

Degree & Branch	M. Tech Computer Science & Engineering	Semester	VI
Subject Code & Name	ICS1512 – Machine Learning Algorithms Laboratory		
Academic Year	2026–2027 (odd)	Batch	2024–2029
Due Date			

Experiment 3: Regression Analysis using Linear and Regularized Models

Objective

To implement linear and regularized regression models for predicting a continuous target variable, evaluate their performance using multiple metrics, visualize model behavior, and analyze overfitting, underfitting, and bias–variance characteristics.

Dataset

A real-world regression dataset containing numerical and categorical features related to loan applications is used. The target variable is the **loan amount sanctioned**.

Dataset reference:

- Kaggle: Predict Loan Amount Data

Brief Theory (For Lab Understanding)

Linear Regression

Linear Regression models the relationship between input features and a continuous target variable. It is simple, interpretable, and serves as a baseline regression model.

Regularized Regression Models

Regularization techniques are used to control model complexity:

- Ridge Regression reduces coefficient magnitudes
- Lasso Regression performs feature selection
- Elastic Net combines Ridge and Lasso behavior

Regularization helps improve generalization and reduce overfitting.

Task Description

Students must:

- Implement Linear, Ridge, Lasso, and Elastic Net regression models
- Tune regularization hyperparameters using Grid Search or Randomized Search
- Visualize regression results and errors
- Analyze overfitting, underfitting, and bias–variance trade-off

Implementation Steps

1. Load the dataset
2. Perform data preprocessing:
 - Handle missing values
 - Encode categorical variables
 - Standardize numerical features
3. Perform Exploratory Data Analysis (EDA)
4. Visualize feature distributions and target distribution
5. Split the dataset into training and testing sets
6. Train baseline Linear Regression
7. Train Ridge, Lasso, and Elastic Net models
8. Perform hyperparameter tuning using 5-Fold Cross-Validation
9. Evaluate all models using regression metrics

Required Visualizations (During Coding)

Students must generate and include:

- Target variable distribution plot
- Feature vs. target scatter plots
- Predicted vs. actual values plot
- Residual plot
- Training error vs. validation error plot
- Coefficient comparison bar plot

Performance Metrics to be Reported

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score
- Training Time

Hyperparameter Search Space

- Ridge: $\alpha \in \{0.01, 0.1, 1, 10, 100\}$
- Lasso: $\alpha \in \{0.001, 0.01, 0.1, 1, 10\}$
- Elastic Net:
 - $\alpha \in \{0.01, 0.1, 1, 10\}$
 - $l1_ratio \in \{0.2, 0.5, 0.8\}$

Hyperparameter Tuning Results

Table 1: Hyperparameter Tuning Summary

Model	Search Method	Best Parameters	Best CV R^2
Ridge Regression	Grid / Random		
Lasso Regression	Grid / Random		
Elastic Net Regression	Grid / Random		

Cross-Validation Performance ($K = 5$)

Table 2: Cross-Validation Performance

Model	MAE	MSE	RMSE	R^2
Linear Regression				
Ridge Regression				
Lasso Regression				
Elastic Net Regression				

Table 3: Test Set Performance

Model	MAE	MSE	RMSE	R^2
Linear Regression				
Ridge Regression				
Lasso Regression				
Elastic Net Regression				

Test Set Performance Comparison

Effect of Regularization on Coefficients

Table 4: Coefficient Comparison

Feature	Linear	Ridge	Lasso	Elastic Net
Feature 1				
Feature 2				
Feature 3				

Overfitting and Underfitting Analysis

Students must discuss:

- Difference between training and validation errors
- Effect of regularization strength
- Improvement in generalization after tuning

Bias–Variance Analysis

Students must comment on:

- Bias behavior of Linear Regression
- Variance reduction using Ridge and Elastic Net
- Feature sparsity effect in Lasso

Conclusion

Summarize the performance of all regression models, justify the choice of optimal hyperparameters, and discuss the trade-off between accuracy and model complexity.

Report Format (Mandatory)

1. Aim and Objective
2. Dataset Description
3. Preprocessing Steps
4. Implementation Details
5. Visualizations
6. Performance Tables
7. Overfitting and Underfitting Analysis
8. Bias–Variance Analysis
9. Observations and Conclusion

References

- Scikit-learn: Linear Models
- Scikit-learn: Hyperparameter Optimization
- Loan Amount Dataset